AIX-MARSEILLE UNIVERSITÉ

GROUP WORK

Detection and Identification of Built-up Types Using Hyperspectral Imagery in Toulouse

Hyperspectral analysis of roofing materials and built-up surfaces

GROUP MEMBERS

BAKHOUM Abdourahmane

OMAR Faza

SOUKOU A. Ezechiel

Supervisor: Prof. Sébastien GADAL

Master Geomatics and Spatial Modelling (GMS) 2026–2027

Contents

1. Introduction and objective **3**

2. Data and spectral selection **3**

2.1 Main dataset **3**

2.2 Selected bands **3**

2.3 Target material classes **3**

3. Methodology **4**

3.1 Processing chain **4**

3.2 Role of each step **4**

4. Summary table: materials, bands and processing **5**

4.1 Workflow justification **5**

5. Results **6**

5.1 PCA and MNF components **6**

5.2 Extraction and classification of built-up surfaces **6**

6. Summary **7**

7. Limitations and points of caution **7**

1. Introduction and objective

This work focuses on the detection and identification of the main types of roofing materials from an aerial hyperspectral image of Toulouse. It aims to maintain a coherent processing chain based on four steps: PCA, MNF, masking and K-means classification.

The value of hyperspectral imagery is that it records radiation in a very large number of narrow bands. Two roofs that look identical in a colour image may therefore have different spectral behaviour, particularly in the near-infrared and shortwave infrared. This additional information makes it possible to distinguish materials, not just shapes.

This richness comes at a cost: the data are voluminous, redundant and noisy. The chosen processing chain addresses this problem. The useful bands are selected first, the information is condensed, surfaces of no interest are removed, and then the data are classified. Each step prepares the next.

Resolution	Bands	Components	Classification
2 m	137 / 295 / 298 / 301	PCA 1 + MNF 2	K-means

2. Data and spectral selection

2.1 Main dataset

The main dataset is an aerial hyperspectral image of Toulouse with a spatial resolution of 2 m, projected in RGF93 / CC43 (EPSG:3943). A 2 m resolution is sufficient to isolate most roofs in an urban neighbourhood, but it is borderline for small buildings, where a single pixel may mix several materials.

2.2 Selected bands

Band(s)	Approx. wavelength	Spectral domain
137	≈ 0.9 µm	Near-infrared (NIR)
295, 298, 301	≈ 1.8 µm	Shortwave infrared (SWIR)

The choice of these two domains is justified. The near-infrared provides information on the overall reflectance of the surface and helps separate built-up surfaces from vegetation, which is highly reflective in this domain. The shortwave infrared is more sensitive to the mineral nature of materials, which helps differentiate concrete, tile and slate. Using only four bands avoids processing a full cube while keeping the discriminating information.

2.3 Target material classes

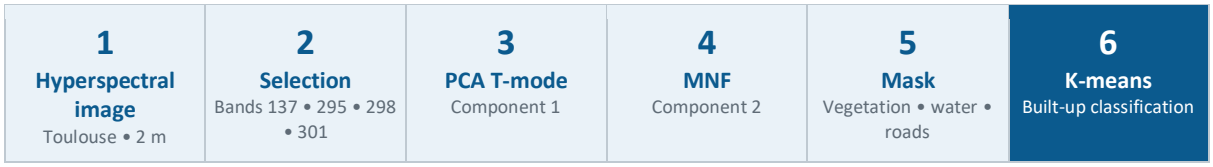
The same band selection and the same processing chain (PCA T-mode → MNF → mask → K-means) apply to all classes.

Class	Target
Concrete	Flat roofs and grey mineral surfaces
Tile	Pitched residential roofs, red/orange covering
Slate	Pitched roofs, dark grey-black covering
Concrete + Slate	Mixed signature of roofing materials
Other	Residual class

3. Methodology

3.1 Processing chain

Bands 137, 295, 298, 301 → PCA T-mode (component 1) → MNF (component 2) → vegetation/water/roads mask → K-means.



3.2 Role of each step

The order of the steps follows a simple logic: dimensionality and noise are reduced first, what is not built-up is then removed, and classification comes last. Classifying before masking would force the algorithm to also separate vegetation, water and roads, at the expense of distinguishing between roofing materials.

- **Spectral selection:** retains the NIR (137) and SWIR (295, 298, 301) bands considered useful for discriminating materials.
- **PCA T-mode (component 1):** summarises the dominant spectral variability of the selected bands.
- **MNF (component 2):** isolates the useful information from noise before masking and classification.
- **Mask:** removes non-built-up pixels (vegetation, water, roads) before classification.
- **K-means:** unsupervised classification of the remaining built-up surfaces.

4. Summary table: materials, bands and processing

The following table specifies, for each object or material, what is detected, the spectral range used, the mathematical composition applied and the classification algorithm.

Object / material	Type of material / surface detected	Selected spectral range	Mathematical composition	Classification algorithm
Concrete	Flat roofs and grey mineral surfaces	NIR ≈ 0.9 µm (band 137) + SWIR ≈ 1.8 µm (bands 295, 298, 301)	PCA T-mode (comp. 1) + MNF (comp. 2)	K-means, after masking
Tile	Pitched residential roofs, red/orange covering	NIR ≈ 0.9 µm (137) + SWIR ≈ 1.8 µm (295, 298, 301)	PCA T-mode (comp. 1) + MNF (comp. 2)	K-means, after masking
Slate	Pitched roofs, dark grey-black covering	NIR ≈ 0.9 µm (137) + SWIR ≈ 1.8 µm (295, 298, 301)	PCA T-mode (comp. 1) + MNF (comp. 2)	K-means, after masking
Concrete + Slate	Mixed signature of roofing materials	NIR ≈ 0.9 µm (137) + SWIR ≈ 1.8 µm (295, 298, 301)	PCA T-mode (comp. 1) + MNF (comp. 2)	K-means, after masking
Vegetation, water, roads (to exclude)	Non-built-up surfaces, removed before classification	To be specified according to the masking method used	To be specified (index or thresholding)	Masking (no classification)

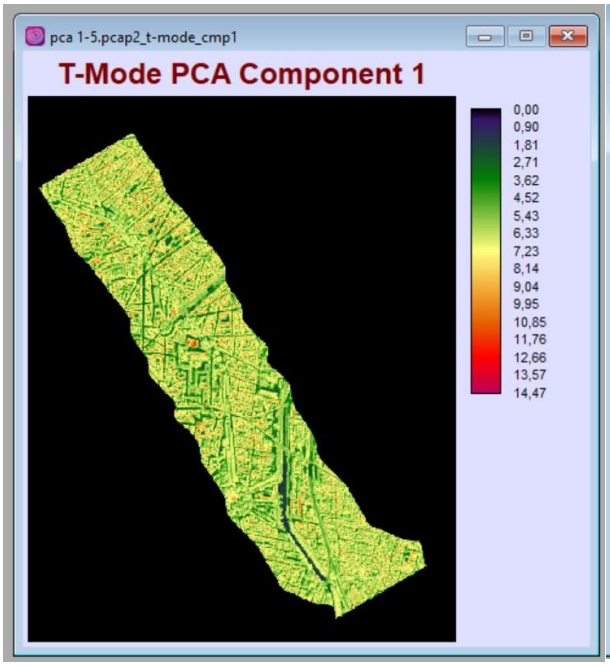
4.1 Workflow justification

Each step meets a specific need, in a logical order: reduce and denoise the information, remove what is not built-up, then classify.

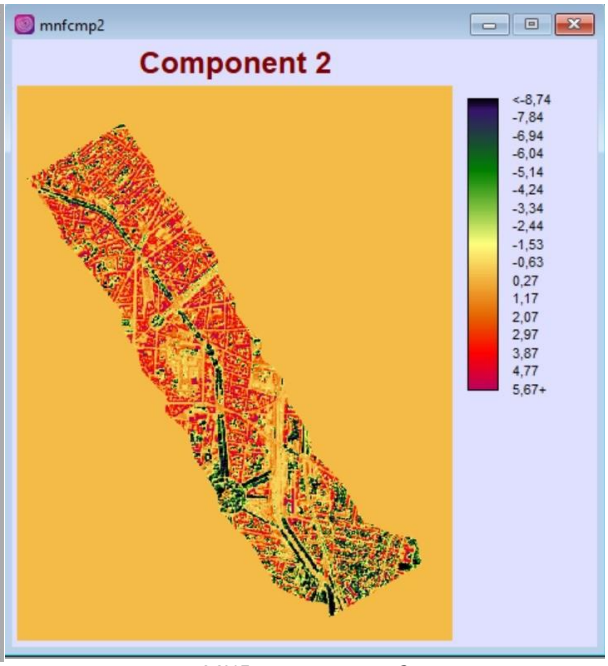
Step	Choice	Why
Band selection	137 (NIR); 295, 298, 301 (SWIR)	NIR and SWIR are the domains where mineral and roofing materials are best distinguished; these bands also limit the redundancy of a very large hyperspectral cube.
PCA T-mode	Component 1	Condenses the dominant spectral variability of the selected bands into synthetic information.
MNF	Component 2	Separates useful information from noise before masking and classification, which makes the clusters more stable.
Mask	Vegetation, water, roads	Removes non-built-up pixels so that K-means does not create clusters on these surfaces and focuses on roofs.
K-means	Unsupervised classification	Requires no training data; suited to grouping similar signatures (Slate, Concrete, Tile, mixed) and isolating an “Other” class.

5. Results

5.1 PCA and MNF components



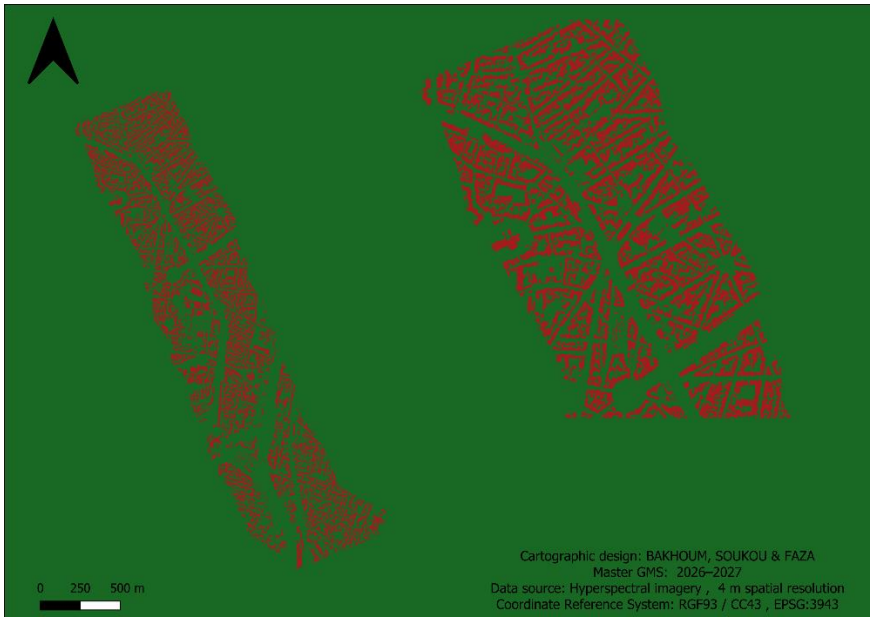
PCA T-mode — component 1



MNF — component 2

Reading: component 1 of the PCA summarises the dominant spectral variability of the selected bands. Component 2 of the MNF is used in the processing chain to better isolate useful information from noise.

5.2 Extraction and classification of built-up surfaces



Extraction of built-up surfaces after masking

The extraction after masking shows that only built-up surfaces are kept, which reduces the number of pixels to classify. The K-means classification map then assigns each remaining pixel to a material class.



K-means classification of built-up surfaces

The mapping results distinguish the Slate, Concrete, Concrete + Slate and Tile classes, as well as a residual "Other" class. This class groups pixels that match none of the retained signatures, for example roofs of a different nature or mixed pixels. Since K-means is unsupervised, the link between a cluster and a material must be confirmed by verification, for example with reference samples or field imagery.

6. Summary

The PCA–MNF combination condenses the selected spectral information before the masking step. The mask removes non-built-up pixels before the K-means classification. The mapping results notably distinguish the Slate, Concrete, Concrete + Slate and Tile classes, with a residual "Other" class.

The proposed chain is coherent because each choice follows from the previous one: the NIR and SWIR bands provide the useful information, PCA and MNF condense and denoise it, the mask focuses the analysis on built-up areas and K-means groups similar signatures. It remains deliberately simple, which makes it reproducible and easy to justify.

7. Limitations and points of caution

- The wavelengths are approximate: check the exact values in the sensor's .hdr file (notably for bands 295, 298 and 301).
- HyperSIG/HyperSAM results are not included in this version.
- Since the classification is unsupervised (K-means), the match between clusters and materials relies on interpretation.